Basic Mathematics

Mathematics

Lecture Notes

A friendly path through algebra, geometry, and calculus.

June 6, 2026

Contents

Chapter 1

Introduction

These notes are an invitation to look at mathematics in a different way. For many students, mathematics has long appeared as a collection of formulas, rules, and mnemonic patterns: one recognizes the type of problem, chooses a remembered procedure, performs the required operations, and moves on. Such skills have their place, but they are not the heart of mathematics. Mathematics begins when we stop for a moment and ask what we are really looking at, what kind of object has appeared, what question is being asked, what is known, what is unknown, and why a proposed step is justified.

The purpose of this course is therefore not only to practise calculations. We will learn how to read mathematical expressions, how to formulate precise questions, how to distinguish an object from its notation, how to notice structure, and how to build simple but meaningful chains of reasoning. A solved problem is not valuable only because it produces an answer. It is valuable because it can reveal a method of thought: how to begin, what to observe, which assumptions matter, what can be transformed, what remains unchanged, and how a conclusion follows from earlier facts.

The course will focus on three fundamental parts of mathematics: linear algebra, analytic geometry, and calculus. Linear algebra introduces vectors, matrices, systems of equations, and transformations. Analytic geometry shows how algebraic equations describe points, lines, planes, directions, distances, and angles. Calculus gives a language for functions, limits, continuity, derivatives, integrals, change, and accumulation. These topics are not isolated chapters placed next to one another. Together they form a basic language for describing structure, space, and change.

We will try to study this language from the inside. A definition should not be treated as a sentence to memorize, but as a way of introducing a new kind of object. A theorem should not be treated as a decorative statement, but as the result of asking a clear question under precise assumptions. An example should not be treated as a template to copy blindly, but as a small experiment in reasoning. Even a simple calculation can show something important if we ask why it works and what it tells us.

For this reason, some arguments will deliberately return in different places. The same habit of asking what is fixed and what is variable appears when we study functions, vectors, equations, and transformations. The same need for precision appears when we define a line, compute a derivative, solve a system of equations, or interpret an integral. The same mathematical question can often be seen algebraically, geometrically, and conceptually. This repetition is intentional: it helps the reader recognize patterns of reasoning rather than only patterns of exercises.

The number of solved tasks is not the main measure of progress. What matters is how the tasks are solved. Did we understand the question? Did we identify the object correctly? Did we choose notation consciously? Did we know why each step was allowed? Did we pause to interpret the result? Did the calculation teach us something about the idea behind it? These questions are part of the work. They are not an addition to mathematics; they are mathematics being done carefully.

Mathematics is useful in physics, engineering, computer science, data analysis, modelling, and many other fields, but its value is not only practical. It is also a way of seeing. It teaches precision, discipline, imagination, and the ability to move from simple observations to general conclusions. These notes are meant to show a piece of that beauty, strength, and richness. The goal is not merely to survive mathematical techniques, but to begin to appreciate mathematics as an art of asking good questions and giving well-reasoned answers.

Chapter 2

Matrices

2.1 What matrices are

A matrix is a rectangular arrangement of mathematical objects, usually numbers. This description is correct, but it is not yet very informative. If we stop at this description, a matrix looks like a table, and matrix algebra looks like a collection of rules for moving numbers around. In this chapter we will try to see more than that. A matrix is a way of organizing information. It can describe data, relations between quantities, coefficients of equations, or a rule that transforms one vector into another.

The first skill is therefore not calculation, but reading. When we see a matrix, we should ask what its shape is, what its entries represent, and how its rows and columns are meant to be interpreted. The same array of numbers may be only a table in one problem, but in another problem it may describe a transformation, a system of equations, or a geometric operation.

Definition

A matrix with m rows and n columns is called an $m \times n$ matrix. It has the form

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

The entry a_{ij} is located in row i and column j.

This notation is simple, but it contains an important discipline. The first index tells us where to look vertically, and the second index tells us where to look horizontally. A student who confuses these indices will often make mistakes later when multiplying matrices or describing systems of equations.

Example

Consider the matrix

$$M = \begin{pmatrix} 4 & -2 & 0 \\ 7 & 1 & 5 \end{pmatrix}.$$

This is a 2×3 matrix. The entry m_{11} is 4, the entry m_{12} is -2, and the entry m_{23} is 5. We are not doing any operation yet. We are only learning to read the object.

The size of a matrix is part of its identity. A row matrix, a column matrix, and a square matrix may contain similar numbers, but they play different roles. A column matrix can represent a vector. A square matrix can sometimes represent a transformation from a space to itself. A rectangular matrix may describe a relation between two collections of quantities of different sizes.

Definition

A matrix is called square if it has the same number of rows and columns. An $n \times n$ matrix is a square matrix of order n.

We will often begin with square matrices because many basic ideas are easiest to see there. However, rectangular matrices are just as important. The point is not to memorize a list of shapes, but to develop the habit of asking: what kind of object am I holding, and what operations make sense for this object?

2.2 Basic matrix operations

Once we know how to read a matrix, we can begin to operate on it. The simplest operations are addition, subtraction, and multiplication by a number. These operations are deliberately simple because they preserve the shape of the matrix. They do not change a table into a different kind of object; they change its entries while keeping the same positions.

Definition

Let $A = (a_{ij})$ and $B = (b_{ij})$ be matrices of the same size $m \times n$. Their sum is the matrix

$$A + B = (a_{ij} + b_{ij}),$$

and their difference is the matrix

$$A - B = (a_{ij} - b_{ij}).$$

If λ is a number, then

$$\lambda A = (\lambda a_{ij}).$$

The phrase "of the same size" is not a technical detail. It is the reason the operation is meaningful. When two matrices have the same size, each entry of one matrix has a partner in the other matrix. Addition and subtraction then happen position by position.

Example

Let

$$A = \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 5 & 4 \\ -2 & 1 \end{pmatrix}.$$

Then

$$A + B = \begin{pmatrix} 2+5 & -1+4 \\ 0+(-2) & 3+1 \end{pmatrix} = \begin{pmatrix} 7 & 3 \\ -2 & 4 \end{pmatrix}.$$

Also,

$$A - B = \begin{pmatrix} 2-5 & -1-4 \\ 0-(-2) & 3-1 \end{pmatrix} = \begin{pmatrix} -3 & -5 \\ 2 & 2 \end{pmatrix}.$$

Finally,

$$-3A = \begin{pmatrix} -6 & 3 \\ 0 & -9 \end{pmatrix}.$$

The calculation is not difficult. The important thing is the discipline of positions. We do not add rows to columns, and we do not rearrange entries because it looks convenient. We respect the shape of the object.

These operations satisfy the familiar rules one expects from ordinary arithmetic. For matrices of the same size,

$$A + B = B + A, \qquad (A + B) + C = A + (B + C).$$

This should not be surprising: addition is performed entry by entry, and ordinary addition of numbers is commutative and associative.

Remark

Whenever a matrix operation is defined entry by entry, many properties of ordinary arithmetic pass directly to matrices. But this principle must be used carefully. Matrix multiplication will not be entry-by-entry multiplication, and therefore it will behave differently.

2.3 Matrix multiplication

Matrix multiplication is the first operation in this chapter that is not simply performed position by position. This is why it deserves careful attention. Many mistakes with matrices come from treating multiplication as if it were only another version of addition. It is not. Addition compares matching positions. Multiplication combines rows of the first matrix with columns of the second matrix.

Definition

Let $A = (a_{ij})$ be an $m \times n$ matrix and let $B = (b_{ij})$ be an $n \times p$ matrix. The product AB is the $m \times p$ matrix $C = (c_{ij})$, where

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

In words, the entry c_{ij} is obtained by multiplying row i of A by column j of B.

The condition on sizes is built into the definition. If A has size $m \times n$, then each row of A has n entries. To multiply this row by a column of B, each column of B must also have n entries. Therefore B must have n rows. We summarize this by writing

$$(m \times n)(n \times p) = m \times p.$$

The inner dimensions must match; the outer dimensions describe the size of the result.

Example

Let

$$A = \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 0 \\ 5 & -1 \end{pmatrix}.$$

The first entry of AB comes from the first row of A and the first column of B:

$$1 \cdot 2 + 3 \cdot 5 = 17.$$

The first row and second column give

$$1 \cdot 0 + 3 \cdot (-1) = -3.$$

Continuing in the same way,

$$AB = \begin{pmatrix} 17 & -3 \\ 16 & -4 \end{pmatrix}.$$

The example is small, but the point is general. Each entry of the product answers one local question: how does a chosen row of the first matrix interact with a chosen column of the second matrix?

Why the order matters

For ordinary numbers, $ab = ba$. For matrices this is not generally true. The reason is conceptual, not only computational. The product AB asks rows of A to interact with columns of B. The product BA asks a different question.

Using the matrices from the previous example,

$$BA = \begin{pmatrix} 2 & 0 \\ 5 & -1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 6 \\ 7 & 11 \end{pmatrix}.$$

Thus $AB \neq BA$. The matrices have not changed, but the order has changed, and the question asked by the multiplication has changed.

Remark

The statement $AB \neq BA$ does not mean that two matrices can never commute. Some matrices do commute. The point is that commutativity is not automatic. If it matters, it must be checked or justified from the structure of the matrices.

Matrix multiplication is associative:

$$(AB)C = A(BC),$$

whenever the products are defined. This means that if several compatible matrices are multiplied in a fixed order, parentheses do not change the result. But the order of the matrices themselves still matters. Associativity tells us how to group operations. It does not allow us to swap them.

2.4 Special matrices and visible structure

Some matrices are important because their structure is easy to see. They help us learn a useful lesson: a matrix is not only a collection of entries. The position of the entries matters, and certain patterns can make an operation simple or meaningful.

The zero matrix and the identity matrix

The zero matrix is the matrix whose entries are all zero. It behaves like zero for matrix addition. If A is an $m \times n$ matrix and 0 is the zero matrix of the same size, then

$$A + 0 = A.$$

The identity matrix plays a similar role for multiplication, but only for square matrices.

Definition

The $n \times n$ identity matrix is

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

For compatible matrices,

$$I_n A = A, \qquad AI_n = A.$$

The identity matrix is important because it represents doing nothing. This phrase is informal, but helpful. Multiplying by the identity preserves the object, just as multiplying a number by 1 preserves the number.

Diagonal matrices

A diagonal matrix is a square matrix whose non-zero entries, if any, lie only on the main diagonal.

Definition

A square matrix $D = (d_{ij})$ is diagonal if

$$d_{ij} = 0 \quad \text{whenever } i \neq j.$$

We often write

$$D = \operatorname{diag}(d_1, d_2, \ldots, d_n).$$

Diagonal matrices are useful because they act independently on different positions. There is no mixing between coordinates. This is why many calculations with diagonal matrices are transparent.

Example

Let

$$D = \operatorname{diag}(3, -1, 2).$$

Then

$$D^2 = \operatorname{diag}(9, 1, 4), \qquad D^3 = \operatorname{diag}(27, -1, 8).$$

The power is computed by taking powers of the diagonal entries. This works because all off-diagonal entries are zero, so the coordinates do not mix.

If two diagonal matrices have the same size, then they commute. Indeed,

$$\operatorname{diag}(a_1, \ldots, a_n) \operatorname{diag}(b_1, \ldots, b_n) = \operatorname{diag}(a_1 b_1, \ldots, a_n b_n),$$

and the same result is obtained in the opposite order because ordinary numbers commute. This is a good example of structure explaining an algebraic fact.

Patterns worth noticing

When a matrix appears in a problem, it is useful to ask whether it has visible structure. Is it diagonal? Does it contain many zeros? Are two rows equal? Are rows or columns multiples of one another? Does it look symmetric? Such observations can simplify computations, but more importantly they help us understand the object.

Remark

A calculation often becomes easier after we notice structure. But noticing structure is not a trick; it is part of mathematical understanding. We learn not only to compute with objects, but also to see them.

2.5 Row operations and problem solving

Matrices are often used because they allow us to transform a problem without rewriting every sentence of the problem. One of the most important examples is the use of row operations. Row operations will become essential when we solve systems of linear equations, but it is useful to meet them already here.

A row operation changes a matrix in a controlled way. We do not change entries randomly. We perform an operation that has a clear description and a clear purpose.

Definition

The basic elementary row operations are:

- swapping two rows,
- multiplying a row by a non-zero number,
- adding a multiple of one row to another row.

The value of row operations is not that they make matrices look different. Their value is that they help reveal structure. For example, they can help us simplify a matrix, detect whether equations are dependent, or prepare a system for solving.

Example

Consider the matrix

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & 3 \\ 0 & 1 & 5 \end{pmatrix}.$$

Suppose we replace the second row by the second row minus twice the first row. Then

$$(2, 4, 3) - 2(1, 2, -1) = (0, 0, 5).$$

The new matrix is

$$\begin{pmatrix} 1 & 2 & -1 \\ 0 & 0 & 5 \\ 0 & 1 & 5 \end{pmatrix}.$$

This operation did not merely produce new numbers. It revealed that part of the second row was already explained by the first row, except for the last entry.

This example should be read slowly. The operation has three layers. First, there is arithmetic. Second, there is a transformation of the row. Third, there is interpretation: the new row tells us something about dependence between rows.

Column vectors as matrices

A column vector may be viewed as a matrix with one column. This viewpoint will be used constantly in linear algebra. It allows us to write vector equations and matrix equations in the same language.

For instance,

$$u = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$$

is a 3×1 matrix. Its transpose is the row matrix

$$u^\top = \begin{pmatrix} 2 & -1 & 4 \end{pmatrix}.$$

The distinction between a row and a column is not cosmetic. The products $u^\top u$ and $uu^\top$ have different sizes and different meanings:

$$u^\top u$$

is a 1×1 matrix, while

$$uu^\top$$

is a 3×3 matrix.

This is another example of a recurring principle: before calculating, read the shapes.

How to approach matrix exercises

When solving matrix exercises, the aim is not to rush toward the answer. The answer is only the last line of a process. A good solution should make the process visible.

Summary

When working with matrices, ask:

- What is the size of each matrix?
- Which operations are defined?
- Am I adding entries position by position, or multiplying rows by columns?
- Does the order of multiplication matter here?
- Is there a visible structure that simplifies the problem?
- What does the result tell me about the original objects?

A student who asks these questions is not merely applying a recipe. They are beginning to use matrices as mathematical objects. That is the real purpose of this chapter.

Chapter 3

Determinants

3.1 Why determinants matter

In the previous chapter we learned how to read matrices and how to perform basic operations on them. We can add matrices, multiply them by numbers, multiply compatible matrices, and recognize simple structures such as the identity matrix or a diagonal matrix. But after learning these operations, a natural question appears.

Are all square matrices equally good?

This question is intentionally vague at first. Mathematics often begins in this way: we notice that some objects behave differently from others, and only later do we find the right words and the right definitions. Consider the following matrices:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

The first matrix is the identity matrix. It preserves everything under multiplication. The second matrix stretches the first coordinate by a factor of 2 and the second coordinate by a factor of 3. The third matrix is different. Its second row is exactly twice its first row:

$$(2, 4) = 2(1, 2).$$

So the two rows do not contain two independent pieces of information. One row is only a repetition of the other, scaled by a number.

This observation suggests a deeper question. Can a matrix lose information? Can a matrix collapse different inputs into outputs that are no longer distinguishable? Can we detect this collapse by assigning one carefully chosen number to the matrix?

The determinant is such a number.

This statement should not be read as a definition yet. At this stage it is only a promise. We are looking for a number attached to a square matrix, but not just any number. We want a number that reacts to the structure of the matrix. It should notice when rows or columns become dependent. It should distinguish matrices that preserve two-dimensional or three-dimensional structure from matrices that collapse it. It should also help us decide, later, whether a matrix can be inverted.

Remark

A determinant is not a summary of all entries of a matrix in the sense of an average or a sum. It is a number designed to detect structural behaviour. In particular, the value 0 will become extremely important: it will mean that the matrix has lost a certain kind of independence.

There is also a geometric way to think about determinants. A 2×2 matrix can transform the plane. Some matrices stretch areas, some reverse orientation, and some flatten the plane onto a line. The determinant measures this behaviour. In dimension 2, its absolute value describes an area scale factor. In dimension 3, its absolute value describes a volume scale factor. A determinant equal to zero means that area or volume has been flattened completely.

For the purposes of this course, however, we will not begin from geometry alone. We will move between several viewpoints:

- determinants as numbers computed from square matrices,
- determinants as tests for structural collapse,
- determinants as tools for recognizing invertibility,
- determinants as quantities affected in predictable ways by row operations.

The chapter will therefore follow a path of discovery. First we study small matrices, where the formulas can be seen directly. Then we introduce a systematic method for larger matrices: minors, cofactors, and Laplace expansion. After that we collect the properties that make determinants useful in actual calculation. Finally we connect determinants with singular matrices and prepare the transition to the next chapter, where invertibility becomes the central topic.

3.2 Determinants of small matrices

The determinant is defined only for square matrices. This is already meaningful: we are not attaching this number to every rectangular table. We are studying matrices that can be interpreted as transforming a space into itself, or as square coefficient matrices in systems of equations.

We begin with small cases. This is not only because small cases are easier. Small cases allow us to see the idea before it becomes hidden behind notation.

3.2.1 The determinant of a 1×1 matrix

The simplest square matrix has only one entry:

$$A = \begin{pmatrix} a \end{pmatrix}.$$

There is no structure to compare, no rows to exchange, and no area or volume to measure. The determinant is simply the entry itself:

$$\det A = a.$$

This case looks trivial, but it is useful because larger definitions will reduce determinants to smaller determinants. Eventually, a long computation ends at 1×1 determinants.

3.2.2 The determinant of a 2×2 matrix

Now consider a general 2×2 matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

The determinant is defined by

$$\det A = ad - bc.$$

The formula should be read carefully. We multiply along the main diagonal and subtract the product along the other diagonal:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

At first this may look like an arbitrary rule. It is not arbitrary. It is a rule that reacts very strongly to dependence between rows or columns.

Example

For

$$A = \begin{pmatrix} 3 & 1 \\ 2 & 5 \end{pmatrix}$$

we get

$$\det A = 3 \cdot 5 - 1 \cdot 2 = 15 - 2 = 13.$$

The determinant is non-zero. This suggests that the two rows are not repetitions of each other and that the matrix has not collapsed its two-dimensional structure.

Example

For

$$B = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}$$

we get

$$\det B = 1 \cdot 6 - 2 \cdot 3 = 6 - 6 = 0.$$

The second row is three times the first row:

$$(3, 6) = 3(1, 2).$$

Thus the determinant detects that the two rows are dependent.

This example gives us the first important interpretation: when the determinant of a 2×2 matrix is zero, the two rows or the two columns fail to provide two independent directions of information.

3.2.3 What changes the determinant?

Let us observe how the formula reacts to simple operations. For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

we have

$$\det A = ad - bc.$$

If we exchange the two rows, we obtain

$$A' = \begin{pmatrix} c & d \\ a & b \end{pmatrix}.$$

Then

$$\det A' = cb - da = -(ad - bc) = -\det A.$$

So exchanging two rows changes the sign of the determinant.

If the two rows are equal,

$$A = \begin{pmatrix} a & b \\ a & b \end{pmatrix},$$

then

$$\det A = ab - ba = 0.$$

This is exactly what we should expect: two equal rows certainly do not provide two independent directions.

Remark

The determinant is sensitive to order and independence. It changes sign when rows are swapped, and it becomes zero when rows become dependent. These facts will later become general properties for determinants of all sizes.

3.2.4 The determinant of a 3×3 matrix and Sarrus' rule

For a 3×3 matrix

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix},$$

one useful formula is

$$\det A = aei + bfg + cdh - ceg - bdi - afh.$$

This formula can be remembered using Sarrus' rule. Repeat the first two columns after the matrix:

$$\begin{array}{ccc|cc} a & b & c & a & b \\ d & e & f & d & e \\ g & h & i & g & h \end{array}$$

Then add the products along the three downward diagonals and subtract the products along the three upward diagonals:

$$\det A = (aei + bfg + cdh) - (ceg + bdi + afh).$$

Example

Let

$$A = \begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 2 & 0 & 5 \end{pmatrix}.$$

Using Sarrus' rule,

$$\det A = (1 \cdot 3 \cdot 5 + 2 \cdot 4 \cdot 2 + 0 \cdot (-1) \cdot 0) - (0 \cdot 3 \cdot 2 + 2 \cdot (-1) \cdot 5 + 1 \cdot 4 \cdot 0).$$

Hence

$$\det A = (15 + 16 + 0) - (0 - 10 + 0) = 31 - (-10) = 41.$$

The determinant is non-zero.

The computation above is mechanical, but it is still worth reading. A non-zero determinant tells us that the rows and columns of the matrix do not collapse into a lower-dimensional structure. Later this will imply that the matrix is invertible.

Remark

Sarrus' rule is convenient, but it is special. It works for 3×3 matrices only. It is not a method for 4×4 or larger matrices. For larger matrices we need a more general idea.

3.2.5 A warning about calculation

When computing determinants, students often try to memorize many separate rules. A better habit is to ask three questions:

- What is the size of the matrix?

- Which method is appropriate for this size and this structure?
- Is there visible structure that can simplify the calculation?

For 2×2 matrices, the formula $ad - bc$ is usually fastest. For 3×3 matrices, Sarrus' rule may be useful. But as soon as the matrix becomes larger, or as soon as many zeros appear, the method of expansion by minors and cofactors becomes much more important.

3.3 Minors, cofactors, and Laplace expansion

Sarrus' rule gives a quick method for 3×3 matrices, but it does not generalize to larger matrices. If we want a method that works for every square matrix, we need a different idea. The key idea is reduction: we compute a large determinant by reducing it to smaller determinants.

This is a recurring mathematical pattern. When an object is too large to understand at once, we try to break it into smaller objects of the same kind. For determinants, the smaller objects are called minors.

3.3.1 Minors

Let $A = (a_{ij})$ be an $n \times n$ matrix. Choose an entry a_{ij}. Remove row i and column j. The determinant of the remaining $(n-1) \times (n-1)$ matrix is called the minor of a_{ij}, and is denoted by M_{ij}.

Example

Let

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 5 & 2 & 1 \end{pmatrix}.$$

To find M_{12}, we remove row 1 and column 2. The remaining matrix is

$$\begin{pmatrix} 0 & 4 \\ 5 & 1 \end{pmatrix}.$$

Therefore

$$M_{12} = \det \begin{pmatrix} 0 & 4 \\ 5 & 1 \end{pmatrix} = 0 \cdot 1 - 4 \cdot 5 = -20.$$

A minor is not an entry of the original matrix. It is a determinant obtained after deleting a row and a column. This distinction matters: the entry a_{ij} is a number already present in the matrix; the minor M_{ij} is computed from what remains after removing the row and column of that entry.

3.3.2 Cofactors

The minor still needs a sign. The signs follow a checkerboard pattern:

$$\begin{pmatrix} + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

This pattern is encoded by the factor $(-1)^{i+j}$.

Definition

The cofactor of the entry a_{ij} is

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

Example

Using the matrix from the previous example, we found

$$M_{12} = -20.$$

Since $(-1)^{1+2} = (-1)^3 = -1$, the cofactor is

$$C_{12} = (-1)M_{12} = 20.$$

For the entry in position $(1, 1)$, the sign is positive. Removing row 1 and column 1, we get

$$M_{11} = \det \begin{pmatrix} -1 & 4 \\ 2 & 1 \end{pmatrix} = (-1) \cdot 1 - 4 \cdot 2 = -9.$$

Thus

$$C_{11} = M_{11} = -9.$$

The alternating signs are not decoration. Without them, the determinant would not have the correct behaviour under row exchange, and it would not detect orientation properly.

3.3.3 Laplace expansion along a row

Laplace expansion expresses a determinant as a sum of entries multiplied by their cofactors. If we expand along row i, then

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}.$$

Equivalently,

$$\det A = \sum_{j=1}^{n} a_{ij}C_{ij}.$$

Example

Let

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 5 & 2 & 1 \end{pmatrix}.$$

Expand along the first row. We need the cofactors C_{11}, C_{12}, and C_{13}.
First,

$$C_{11} = +\det \begin{pmatrix} -1 & 4 \\ 2 & 1 \end{pmatrix} = (-1) \cdot 1 - 4 \cdot 2 = -9.$$

Second,

$$C_{12} = -\det \begin{pmatrix} 0 & 4 \\ 5 & 1 \end{pmatrix} = -(0 \cdot 1 - 4 \cdot 5) = 20.$$

Third,

$$C_{13} = +\det \begin{pmatrix} 0 & -1 \\ 5 & 2 \end{pmatrix} = 0 \cdot 2 - (-1) \cdot 5 = 5.$$

Therefore

$$\det A = 2C_{11} + 1C_{12} + 3C_{13} = 2(-9) + 1(20) + 3(5).$$

Hence

$$\det A = -18 + 20 + 15 = 17.$$

Notice how the computation works. A 3×3 determinant was reduced to three 2×2 determinants. For a 4×4 determinant, Laplace expansion reduces the problem to 3×3 determinants, and so on. The method is recursive.

3.3.4 Laplace expansion along a column

We may also expand along a column. If we expand along column j, then

$$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}.$$

Equivalently,

$$\det A = \sum_{i=1}^{n} a_{ij}C_{ij}.$$

The freedom to choose a row or a column is important. A wise choice can make a determinant much easier to compute.

Example

Consider

$$B = \begin{pmatrix} 4 & 0 & 1 & 2 \\ 0 & 3 & 0 & -1 \\ 2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 6 \end{pmatrix}.$$

The second column contains three zeros:

$$\begin{pmatrix} 0 \\ 3 \\ 0 \\ 0 \end{pmatrix}.$$

So we expand along the second column. Only one term survives:

$$\det B = 3C_{22}.$$

Since $(-1)^{2+2} = 1$,

$$C_{22} = \det \begin{pmatrix} 4 & 1 & 2 \\ 2 & 5 & 0 \\ 1 & 0 & 6 \end{pmatrix}.$$

Now compute the remaining 3×3 determinant. Using Sarrus' rule,

$$\det \begin{pmatrix} 4 & 1 & 2 \\ 2 & 5 & 0 \\ 1 & 0 & 6 \end{pmatrix} = (4 \cdot 5 \cdot 6 + 1 \cdot 0 \cdot 1 + 2 \cdot 2 \cdot 0) - (2 \cdot 5 \cdot 1 + 1 \cdot 2 \cdot 6 + 4 \cdot 0 \cdot 0).$$

Thus

$$C_{22} = 120 - (10 + 12) = 98.$$

Therefore

$$\det B = 3 \cdot 98 = 294.$$

This example shows why Laplace expansion is not merely a theoretical formula. It gives a strategy. Before computing, inspect the matrix. A row or column with many zeros is usually the best place to expand.

3.3.5 A method, not a ritual

Laplace expansion can be used along any row or any column. The result is always the same determinant. But the amount of work may be very different. The mathematical question is not only "Can I compute it?" but also "How can I compute it intelligently?"

Summary

When using Laplace expansion, follow this order.

- Inspect the matrix before calculating.
- Look for a row or column with many zeros.
- Write the sign pattern carefully.
- Compute the minors as smaller determinants.
- Keep the structure visible; do not compress too many steps at once.

3.4 Properties of determinants

At this point we know how to compute determinants of small matrices, and we know a general method: Laplace expansion. But if we use Laplace expansion blindly, determinant calculations can become long and unpleasant. Mathematics is not only about having a method. It is also about knowing when a method can be simplified.

The determinant has several important properties. These properties are not just facts to memorize. They are tools for reading a matrix before calculating.

3.4.1 Triangular matrices

A square matrix is upper triangular if all entries below the main diagonal are zero. It is lower triangular if all entries above the main diagonal are zero.

For example,

$$U = \begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \\ 0 & 0 & -2 \end{pmatrix}$$

is upper triangular.

Theorem

The determinant of a triangular matrix is the product of its diagonal entries.

$$\det U = u_{11}u_{22}\cdots u_{nn}.$$

For the matrix above,

$$\det U = 2 \cdot 3 \cdot (-2) = -12.$$

This is much faster than expanding the determinant directly.

The reason is visible from Laplace expansion. If we expand a triangular matrix along a row or column with many zeros, the determinant keeps reducing to smaller triangular matrices.

Eventually only diagonal entries remain.

3.4.2 Swapping two rows

If two rows of a matrix are exchanged, the determinant changes sign.

Theorem

If B is obtained from A by swapping two rows, then

$$\det B = -\det A.$$

The same is true for swapping two columns.

For a 2×2 matrix this can be checked directly:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc,$$

while

$$\det \begin{pmatrix} c & d \\ a & b \end{pmatrix} = cb - da = -(ad - bc).$$

The general case follows from the same alternating nature of the determinant.

This property explains why the sign pattern in cofactors matters. The determinant remembers orientation. Exchanging two rows reverses that orientation.

3.4.3 Equal rows and proportional rows

If two rows of a matrix are equal, then the determinant is zero.

Indeed, suppose two rows are equal. If we swap these two rows, the matrix does not change. Therefore the determinant should remain the same. But swapping two rows changes the sign of the determinant. Hence

$$\det A = -\det A,$$

so

$$2 \det A = 0,$$

and therefore

$$\det A = 0.$$

This is a good example of a short argument. We did not compute the determinant. We used a property to force its value.

If two rows are proportional, the determinant is also zero. For example,

$$A = \begin{pmatrix} 1 & 3 & -2 \\ 2 & 6 & -4 \\ 0 & 1 & 5 \end{pmatrix}$$

has the second row equal to twice the first row. The determinant is zero because the first two rows do not contain independent information.

Remark

A determinant equal to zero often means that some hidden dependence is present. Repeated rows and proportional rows are the easiest visible cases of this phenomenon.

3.4.4 Multiplying a row by a number

If one row of a matrix is multiplied by a number λ, then the determinant is also multiplied by λ.

Theorem

If B is obtained from A by multiplying one row by λ, then

$$\det B = \lambda \det A.$$

The same is true for multiplying one column by λ.

For example,

$$\det \begin{pmatrix} 2 & 4 \\ 3 & 5 \end{pmatrix} = 2 \cdot 5 - 4 \cdot 3 = 10 - 12 = -2.$$

If we multiply the first row by 3, we get

$$\det \begin{pmatrix} 6 & 12 \\ 3 & 5 \end{pmatrix} = 6 \cdot 5 - 12 \cdot 3 = 30 - 36 = -6.$$

The new determinant is three times the old one:

$$-6 = 3(-2).$$

This property is often used when simplifying determinants or when factoring common numbers out of rows or columns.

3.4.5 Adding a multiple of one row to another

One of the most useful properties is the following.

Theorem

Adding a multiple of one row to another row does not change the determinant.

This property is extremely important because it allows us to use row operations to simplify a determinant without changing its value. It is the determinant version of a powerful idea: transform the object into a simpler one while preserving the quantity we care about.

Example

Compute

$$\det \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 0 & 4 \end{pmatrix}.$$

We simplify the matrix using row operations that do not change the determinant. Replace

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 - R_1.$$

Then

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 0 & 4 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & -2 & 3 \end{pmatrix}.$$

Now replace

$$R_3 \leftarrow R_3 + 2R_2.$$

We get

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 5 \end{pmatrix}.$$

This matrix is upper triangular, so its determinant is the product of the diagonal entries:

$$\det A = 1 \cdot 1 \cdot 5 = 5.$$

This calculation is much more efficient than expanding immediately. It also demonstrates how determinant properties support intelligent computation.

3.4.6 The determinant of a product

Another fundamental property is

$$\det(AB) = \det(A)\det(B),$$

for square matrices A and B of the same size.

This formula says that determinants multiply under matrix multiplication. If one matrix scales area or volume by one factor and another matrix scales it by another factor, then doing both transformations multiplies the scale factors.

One important consequence is immediate. If A has determinant zero, then for every compatible square matrix B,

$$\det(AB) = 0.$$

A matrix that has already collapsed structure cannot be repaired by multiplying by another matrix on one side.

3.4.7 How to use the properties

The properties of determinants are not separate tricks. They form a strategy.

Summary

Before computing a determinant, ask:

- Is the matrix triangular?
- Are two rows or columns equal or proportional?
- Can I create zeros using row operations that preserve the determinant?
- If I swap rows, did I change the sign?
- If I multiply a row by a number, did I multiply the determinant by that number?
- Is Laplace expansion now easier after simplification?

A determinant should not be expanded mechanically unless there is no better path. Often the best computation begins with looking at the matrix and asking what structure is already visible.

3.5 Singularity and parameters

The determinant is especially important because of what happens when it is equal to zero. A zero determinant is not just a numerical accident. It tells us that the matrix has lost a certain kind of independence.

Definition

A square matrix A is called singular if

$$\det A = 0.$$

It is called non-singular if

$$\det A \neq 0.$$

The word "singular" means exceptional or special. In this context, it means that the matrix does not behave like a fully independent square matrix. Something has collapsed. Rows or columns may be dependent, the associated transformation may flatten space, or a system of equations may fail to have a unique solution.

3.5.1 Reading singularity from rows

Consider

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & -1 \end{pmatrix}.$$

The second row is twice the first row:

$$(2, 4, 6) = 2(1, 2, 3).$$

Therefore the rows are dependent, and the determinant is zero:

$$\det A = 0.$$

We do not need to expand the determinant to know this. The structure is visible.

This is an important habit: before calculating, inspect. Sometimes the answer is already present in the shape of the object.

3.5.2 A determinant as a test for invertibility

A central theorem, which will become essential in the next chapter, is the following.

Theorem

A square matrix A is invertible if and only if

$$\det A \neq 0.$$

Equivalently, A is non-invertible if and only if

$$\det A = 0.$$

This theorem is one of the reasons determinants matter. It says that a single number can answer a structural question: can the matrix be reversed? If the determinant is non-zero, the matrix has not collapsed its structure, and an inverse matrix exists. If the determinant is zero, the matrix has lost information, and no inverse matrix can recover it.

We will study inverse matrices in the next chapter. Here we should understand the direction of thought:

$$\det A \neq 0 \quad \Longrightarrow \quad \text{the matrix can be inverted,}$$

while

$$\det A = 0 \quad \Longrightarrow \quad \text{the matrix cannot be inverted.}$$

3.5.3 Matrices depending on a parameter

Determinants are often used when a matrix contains a parameter. Then the determinant becomes an expression depending on that parameter. Solving

$$\det A = 0$$

tells us for which parameter values the matrix becomes singular.

Example

Let

$$A(t) = \begin{pmatrix} t & 1 \\ 4 & t \end{pmatrix}.$$

Then

$$\det A(t) = t \cdot t - 1 \cdot 4 = t^2 - 4.$$

The matrix is singular when

$$t^2 - 4 = 0.$$

Therefore

$$t = 2 \quad \text{or} \quad t = -2.$$

For all other values of t, the matrix is non-singular.

This example is simple, but it shows the main idea. A parameter does not merely make the computation longer. It changes the question. We are no longer asking for one determinant value. We are asking when the structure of the matrix changes.

Example

Let

$$B(a) = \begin{pmatrix} 1 & a & 0 \\ 0 & 1 & a \\ a & 0 & 1 \end{pmatrix}.$$

Using Sarrus' rule,

$$\det B(a) = (1 \cdot 1 \cdot 1 + a \cdot a \cdot a + 0 \cdot 0 \cdot 0) - (0 \cdot 1 \cdot a + a \cdot 0 \cdot 1 + 1 \cdot a \cdot 0).$$

Thus

$$\det B(a) = 1 + a^3.$$

The matrix is singular when

$$1 + a^3 = 0,$$

so

$$a = -1.$$

Thus $B(a)$ is singular only for $a = -1$, and non-singular for all other real values of a.

The determinant turns a structural question about a matrix into an algebraic equation. This is one of the most useful aspects of the concept.

3.5.4 What should be concluded?

When a problem asks for parameter values for which a matrix is singular, the answer is not just a list of numbers. The reasoning has a structure:

1. compute $\det A$ as a function of the parameter,
2. solve the equation $\det A = 0$,
3. interpret the roots as values where the matrix loses invertibility.

Summary

The determinant separates square matrices into two broad classes.

- If $\det A \neq 0$, the matrix is non-singular and invertible.
- If $\det A = 0$, the matrix is singular and not invertible.
- If A depends on a parameter, the equation $\det A = 0$ identifies the exceptional parameter values.

This prepares the next chapter. Once we know how to detect whether a matrix is invertible, we can ask a new question: if the inverse exists, how do we find it?

3.6 Determinant patterns

A student who has just learned determinants often tries to compute every determinant by direct expansion. This is understandable, but it is not the best habit. Determinants reward attention to structure. Before expanding, we should ask what kind of matrix we are looking at.

In this final section we collect several patterns that help with determinant calculations. The purpose is not to create a list of tricks. The purpose is to train the eye.

3.6.1 Zeros are invitations

A row or column with many zeros usually tells us where to expand.

Example

Compute

$$\det \begin{pmatrix} 2 & 0 & 1 & 3 \\ 0 & 0 & 5 & 1 \\ 4 & 0 & -1 & 2 \\ 1 & 7 & 0 & 6 \end{pmatrix}.$$

The second column is

$$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 7 \end{pmatrix}.$$

So expansion along the second column gives only one non-zero term:

$$\det A = 7C_{42}.$$

The sign is

$$(-1)^{4+2} = 1,$$

so

$$\det A = 7 \det \begin{pmatrix} 2 & 1 & 3 \\ 0 & 5 & 1 \\ 4 & -1 & 2 \end{pmatrix}.$$

Now the problem has been reduced to a 3×3 determinant.

The point is not that this computation is finished immediately. The point is that the matrix itself suggested a good first move.

3.6.2 Triangular form is a goal

Because triangular matrices have easy determinants, it is often useful to transform a determinant into triangular form using row operations that preserve the determinant.

Example

Let

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 3 & 7 \\ 3 & 5 & 12 \end{pmatrix}.$$

Use row operations that do not change the determinant:

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 - 3R_1.$$

Then

$$A \longrightarrow \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 2 & 6 \end{pmatrix}.$$

Now use

$$R_3 \leftarrow R_3 - 2R_2.$$

We obtain

$$\begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}.$$

This triangular matrix has determinant

$$1 \cdot 1 \cdot 0 = 0.$$

Therefore $\det A = 0$. The final zero row reveals that the original rows were dependent.

This example is valuable because the calculation and the interpretation agree. We did not merely obtain zero. We saw why zero appeared.

3.6.3 Common factors

If every entry in one row has a common factor, that factor can be taken out of the determinant. The same is true for columns.

For example,

$$\det \begin{pmatrix} 2a & 2b \\ c & d \end{pmatrix} = 2 \det \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

This is often useful when determinants contain parameters or expressions.

Example

Compute

$$\det \begin{pmatrix} 3x & 6x \\ 2 & 5 \end{pmatrix}.$$

The first row has a common factor $3x$:

$$\det\begin{pmatrix}3x & 6x\\ 2 & 5\end{pmatrix} = 3x\det\begin{pmatrix}1 & 2\\ 2 & 5\end{pmatrix}.$$

Now

$$\det\begin{pmatrix}1 & 2\\ 2 & 5\end{pmatrix} = 1\cdot 5 - 2\cdot 2 = 1.$$

Hence

$$\det\begin{pmatrix}3x & 6x\\ 2 & 5\end{pmatrix} = 3x.$$

Factoring is not only a simplification of arithmetic. It may reveal parameter values for which the determinant becomes zero.

3.6.4 Vandermonde-type structure

Some determinants have recognizable patterns. One famous example is the Vandermonde determinant:

$$V = \begin{pmatrix}1 & 1 & 1\\ x & y & z\\ x^2 & y^2 & z^2\end{pmatrix}.$$

Its determinant is

$$\det V = (y-x)(z-x)(z-y).$$

This formula is not something we need to memorize at this stage. What matters is the meaning: the determinant becomes zero when two of the parameters are equal. For example, if $x = y$, then the first two columns are equal:

$$\begin{pmatrix}1\\ x\\ x^2\end{pmatrix} = \begin{pmatrix}1\\ y\\ y^2\end{pmatrix}.$$

Therefore the determinant must be zero. The factor $(y-x)$ records this fact algebraically.

This is a powerful idea. A determinant with parameters often contains factors that describe when rows or columns become dependent.

3.6.5 Looking before expanding

The following checklist is often more valuable than a formula.

Summary

Before computing a determinant, ask:

- Is the matrix triangular or almost triangular?
- Is there a row or column with many zeros?
- Are two rows or columns equal or proportional?
- Can I create zeros using row operations that preserve the determinant?
- Is there a common factor in a row or column?
- If parameters are present, for which values could rows or columns become dependent?

- Which method gives the shortest honest calculation?

The last question is important. We do not want cleverness that hides the reasoning. We want calculations that are both efficient and understandable.

What to remember

The determinant is a number attached to a square matrix, but it is not merely a computational ornament. It detects structural information. It becomes zero when independence is lost. It changes predictably under row and column operations. It can be computed by formulas for small matrices, by Laplace expansion in general, and often more efficiently by using determinant properties.

Most importantly, determinants prepare the next question. If $\det A \neq 0$, then the matrix A is invertible. We can now ask how such an inverse matrix is found and how it can be used. That question leads directly to the next chapter.

Chapter 4

Matrix Inversion

4.1 The meaning of an inverse matrix

In the previous chapter we learned that the determinant can distinguish between singular and non-singular square matrices. The condition

$$\det A \neq 0$$

means that the matrix has not collapsed its structure. This raises a natural question: if a matrix has not collapsed information, can we reverse its action?

This question leads to the idea of an inverse matrix.

For ordinary numbers, the inverse of a non-zero number a is the number a^{-1} such that

$$aa^{-1} = 1.$$

For example, the inverse of 5 is $\frac{1}{5}$, because

$$5 \cdot \frac{1}{5} = 1.$$

The number 1 is special because multiplication by 1 changes nothing. In matrix algebra, the identity matrix plays the same role.

Definition

Let A be a square matrix of order n. A matrix B is called an inverse of A if

$$AB = I_n \quad \text{and} \quad BA = I_n.$$

If such a matrix exists, it is denoted by A^{-1}, and we write

$$AA^{-1} = I_n, \qquad A^{-1}A = I_n.$$

The two equations are both important. Matrix multiplication is not generally commutative, so the statements $AB = I$ and $BA = I$ are not automatically the same piece of information. The inverse must undo the matrix on both sides.

Remark

The notation A^{-1} does not mean $\frac{1}{A}$. Matrices are not numbers, and matrix division is not defined in the same way as numerical division. The symbol A^{-1} means: the matrix which reverses the action of A, if such a matrix exists.

4.1.1 Undoing an operation

The idea of an inverse is the idea of undoing. If a matrix A transforms a vector x into a vector y, then

$$Ax = y.$$

If A^{-1} exists, we can apply it to both sides:

$$A^{-1}Ax = A^{-1}y.$$

Since $A^{-1}A = I$, this becomes

$$x = A^{-1}y.$$

Thus the inverse matrix recovers the original vector from the transformed vector.

This explains why singular matrices cannot have inverses. If a matrix collapses different inputs into the same output, then there is no way to recover the original input uniquely. Information has been lost.

Example

Consider the diagonal matrix

$$D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

Multiplication by D doubles the first coordinate and triples the second coordinate:

$$D \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ 3y \end{pmatrix}.$$

To reverse this action, we should divide the first coordinate by 2 and the second coordinate by 3. Therefore

$$D^{-1} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{pmatrix}.$$

Indeed,

$$DD^{-1} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2.$$

This example is simple because the matrix is diagonal. Each coordinate is changed independently. More general matrices mix coordinates, so their inverses are harder to find. But the meaning stays the same: an inverse matrix reverses the action of the original matrix.

4.1.2 When does an inverse exist?

The determinant gives the fundamental test.

Theorem

A square matrix A is invertible if and only if

$$\det A \neq 0.$$

If $\det A = 0$, then A is singular and has no inverse.

This theorem connects the present chapter with the previous one. Determinants do not merely produce numbers. They answer the structural question of whether a matrix can be reversed.

Summary

The inverse matrix should be understood through three questions:

- What operation does the original matrix perform?

- Has this operation lost information?
- If no information was lost, what matrix reverses the operation?

The rest of this chapter develops methods for finding inverse matrices and for using them correctly.

4.2 Inverses of 2×2 matrices

The first non-trivial case is a 2×2 matrix. It is small enough that the inverse can be written explicitly, but rich enough to show the main ideas: the determinant appears, the order of entries matters, and the inverse exists only under a clear condition.

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

We want to find a matrix

$$A^{-1} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$$

such that

$$AA^{-1} = I_2.$$

Multiplying, we get

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{pmatrix}.$$

For this to be the identity matrix, we need

$$\begin{cases} ap + br = 1, \\ cp + dr = 0, \end{cases} \qquad \begin{cases} aq + bs = 0, \\ cq + ds = 1. \end{cases}$$

Thus finding the inverse is already a problem about solving systems of linear equations. Instead of solving these systems every time, we use the known formula.

Theorem

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

If

$$\det A = ad - bc \neq 0,$$

then

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

If $ad - bc = 0$, then A has no inverse.

The formula has two visible parts. First, the determinant appears in the denominator. This is not accidental: if the determinant is zero, the inverse cannot exist. Second, the matrix

$$\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

is obtained by switching the diagonal entries a, d and changing signs of the off-diagonal entries b, c.

Example

Find the inverse of

$$A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}.$$

First compute the determinant:

$$\det A = 2 \cdot 3 - 1 \cdot 5 = 6 - 5 = 1.$$

Since the determinant is non-zero, the inverse exists. Therefore

$$A^{-1} = \frac{1}{1} \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix} = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}.$$

We should verify the result:

$$AA^{-1} = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix} = \begin{pmatrix} 6-5 & -2+2 \\ 15-15 & -5+6 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Verification is not a decorative step. It helps us remember the meaning of the inverse. We are not only producing a matrix; we are producing a matrix that must multiply with the original one to give the identity.

Example

Consider

$$B = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}.$$

The determinant is

$$\det B = 1 \cdot 6 - 2 \cdot 3 = 6 - 6 = 0.$$

The formula cannot be used because it would require division by zero. More importantly, the inverse does not exist. The second row is three times the first row, so the matrix has lost independent information.

This example shows why the determinant test is not just a technical condition. A zero determinant indicates structural collapse. There is no inverse because there is no unique way to undo the action of the matrix.

4.2.1 A formula should still be read

It is tempting to memorize the 2×2 inverse formula mechanically. But it should be read as a statement with meaning:

$$A^{-1} = \frac{1}{\det A} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The determinant controls whether the inverse exists and how strongly the inverse scales the result. The second matrix rearranges the entries in the only way that makes multiplication return the identity.

Summary

For a 2×2 inverse:

- compute $\det A = ad - bc$,
- if $\det A = 0$, stop: the matrix is not invertible,
- if $\det A \neq 0$, use the inverse formula,
- verify the result by multiplying with the original matrix.

4.3 The Gauss–Jordan method

The formula for a 2×2 inverse is useful, but it does not give a practical method for larger matrices. For a 3×3 matrix, and especially for larger matrices, we need a systematic procedure. The most important procedure is the Gauss–Jordan method.

The idea is simple and powerful. If we want to find A^{-1}, we want to transform A into the identity matrix. Every row operation we apply to A must also be applied to the identity matrix beside it. When the left side becomes I, the right side becomes A^{-1}.

We write this as an augmented matrix:

$$[A \mid I].$$

The goal is

$$[A \mid I] \longrightarrow [I \mid A^{-1}].$$

Remark

The method does not work by magic. Row operations correspond to multiplying by elementary matrices. If a sequence of row operations transforms A into I, then the same sequence transforms I into the matrix that reverses A, namely A^{-1}.

4.3.1 A complete 2×2 example

Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}.$$

We form the augmented matrix

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 7 & 0 & 1 \end{array}\right].$$

Our goal is to turn the left block into the identity matrix.

First eliminate the entry below the first pivot:

$$R_2 \leftarrow R_2 - 3R_1.$$

Then

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & -3 & 1 \end{array}\right].$$

Now eliminate the entry above the second pivot:

$$R_1 \leftarrow R_1 - 2R_2.$$

This gives

$$\left[\begin{array}{cc|cc} 1 & 0 & 7 & -2 \\ 0 & 1 & -3 & 1 \end{array}\right].$$

The left side is now I_2, so

$$A^{-1} = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix}.$$

We verify:

$$\begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix} \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} 7-6 & -2+2 \\ 21-21 & -6+7 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

4.3.2 A complete 3×3 example

Now consider

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}.$$

We start with

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right].$$

Use the first row to eliminate the entry below the first pivot:

$$R_3 \leftarrow R_3 - R_1.$$

Then

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & -1 & 1 & -1 & 0 & 1 \end{array}\right].$$

Use the second row to eliminate the entry below the second pivot:

$$R_3 \leftarrow R_3 + R_2.$$

This gives

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 & 1 & 1 \end{array}\right].$$

Normalize the third row:

$$R_3 \leftarrow \frac{1}{2} R_3.$$

Thus

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array}\right].$$

Now eliminate the entry above the third pivot in row 2:

$$R_2 \leftarrow R_2 - R_3.$$

We get

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array}\right].$$

Finally eliminate the entry above the second pivot in row 1:

$$R_1 \leftarrow R_1 - R_2.$$

Therefore

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array}\right].$$

So

$$A^{-1} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

This example is longer than the 2×2 case, but the logic is the same. We create pivots, eliminate entries below and above them, and keep track of every operation on the identity matrix.

4.3.3 What if the method fails?

The Gauss–Jordan method may fail to produce the identity matrix on the left. This happens precisely when the matrix is singular.

Example

Let

$$B = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

Start with

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{array}\right].$$

Apply

$$R_2 \leftarrow R_2 - 2R_1.$$

We obtain

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{array}\right].$$

The left side cannot become the identity matrix, because the second row of the left block is zero. Thus B has no inverse.

This is not a failure of the method. It is information. The method reveals that the matrix is singular.

Summary

The Gauss–Jordan method for finding A^{-1}:

1. Form the augmented matrix $[A \mid I]$.
2. Use elementary row operations to reduce the left block.
3. If the left block becomes I, the right block is A^{-1}.
4. If the left block cannot become I, then A is not invertible.

4.4 The adjugate method

The Gauss–Jordan method is often the most practical method for finding inverse matrices by hand. There is also another method, built from determinants and cofactors. This method is called the adjugate method.

The adjugate method is important for two reasons. First, it gives an explicit formula for the inverse of a matrix. Second, it shows a deep connection between determinants, cofactors, and invertibility.

4.4.1 The cofactor matrix and the adjugate matrix

Let $A = (a_{ij})$ be an $n \times n$ matrix. For each entry a_{ij}, we can compute its cofactor

$$C_{ij} = (-1)^{i+j} M_{ij},$$

where M_{ij} is the minor obtained by deleting row i and column j.

The matrix of cofactors is

$$C = \begin{pmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{pmatrix}.$$

The adjugate matrix is the transpose of the cofactor matrix.

Definition

The adjugate of a square matrix A, denoted by $\operatorname{adj}(A)$, is defined as

$$\operatorname{adj}(A) = C^{\top},$$

where C is the cofactor matrix of A.

The transpose is essential. The adjugate is not simply the cofactor matrix; it is the cofactor matrix with rows and columns exchanged.

4.4.2 The inverse formula

The central formula is the following.

Theorem

If A is a square matrix and $\det A \neq 0$, then

$$A^{-1} = \frac{1}{\det A} \operatorname{adj}(A).$$

This formula should be read slowly. It says that the inverse exists only when $\det A \neq 0$. It also says that the inverse can be built from cofactors. Thus the same objects used to compute determinants also describe the inverse matrix.

A compact way to express the reason behind the formula is

$$A \operatorname{adj}(A) = (\det A) I.$$

If $\det A \neq 0$, we divide by $\det A$ and obtain

$$A \left(\frac{1}{\det A} \operatorname{adj}(A) \right) = I.$$

So

$$\frac{1}{\det A}\operatorname{adj}(A)$$

is the inverse matrix.

4.4.3 A 3×3 example

Let

$$A = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \\ 0 & 2 & 1 \end{pmatrix}.$$

First compute the determinant. Expanding along the first row,

$$\det A = 1 \det \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix} + 2 \det \begin{pmatrix} -1 & 3 \\ 0 & 2 \end{pmatrix}.$$

The first minor gives

$$3 \cdot 1 - 1 \cdot 2 = 1,$$

and the second minor gives

$$(-1) \cdot 2 - 3 \cdot 0 = -2.$$

Therefore

$$\det A = 1 + 2(-2) = -3.$$

Since $\det A \neq 0$, the inverse exists.

Now compute the cofactors. We list them carefully:

$$C_{11} = \det \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix} = 1,$$

$$C_{12} = -\det \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix} = 1,$$

$$C_{13} = \det \begin{pmatrix} -1 & 3 \\ 0 & 2 \end{pmatrix} = -2,$$

$$C_{21} = -\det \begin{pmatrix} 0 & 2 \\ 2 & 1 \end{pmatrix} = 4,$$

$$C_{22} = \det \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = 1,$$

$$C_{23} = -\det \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = -2,$$

$$C_{31} = \det \begin{pmatrix} 0 & 2 \\ 3 & 1 \end{pmatrix} = -6,$$

$$C_{32} = -\det \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix} = -3,$$

$$C_{33} = \det \begin{pmatrix} 1 & 0 \\ -1 & 3 \end{pmatrix} = 3.$$

Thus the cofactor matrix is

$$C = \begin{pmatrix} 1 & 1 & -2 \\ 4 & 1 & -2 \\ -6 & -3 & 3 \end{pmatrix}.$$

Therefore

$$\operatorname{adj}(A) = C^\top = \begin{pmatrix} 1 & 4 & -6 \\ 1 & 1 & -3 \\ -2 & -2 & 3 \end{pmatrix}.$$

Finally,

$$A^{-1} = \frac{1}{-3} \begin{pmatrix} 1 & 4 & -6 \\ 1 & 1 & -3 \\ -2 & -2 & 3 \end{pmatrix}.$$

So

$$A^{-1} = \begin{pmatrix} -\frac{1}{3} & -\frac{4}{3} & 2 \\ -\frac{1}{3} & -\frac{1}{3} & 1 \\ \frac{2}{3} & \frac{2}{3} & -1 \end{pmatrix}.$$

This method is longer than Gauss–Jordan for this example, but it shows exactly how determinants and cofactors build the inverse.

4.4.4 When should this method be used?

For hand calculations, the adjugate method is convenient for 2×2 matrices and sometimes manageable for 3×3 matrices. For larger matrices it becomes computationally heavy because many cofactors must be computed.

Its conceptual value is very high: it explains why the determinant appears in inverse formulas and why zero determinant prevents invertibility.

Summary

The adjugate method:

1. Compute $\det A$.
2. If $\det A = 0$, the inverse does not exist.
3. Compute all cofactors C_{ij}.
4. Form the cofactor matrix C.
5. Transpose it to obtain $\operatorname{adj}(A)$.
6. Use $A^{-1} = \frac{1}{\det A} \operatorname{adj}(A)$.

4.5 Invertibility and structure

It is possible to treat invertibility as a purely computational question: compute the determinant, or try Gauss–Jordan elimination, and see what happens. This works, but it is not the only way to think. Often the structure of a matrix already tells us something about whether an inverse should exist.

The central rule remains the same:

$$A \text{ is invertible} \iff \det A \neq 0.$$

But before computing the determinant, we should learn to look at the matrix.

4.5.1 Diagonal matrices

A diagonal matrix is one of the clearest cases. Let

$$D = \operatorname{diag}(d_1, d_2, \dots, d_n).$$

Then

$$\det D = d_1 d_2 \cdots d_n.$$

Therefore D is invertible exactly when none of the diagonal entries is zero.

If every $d_i \neq 0$, then

$$D^{-1} = \operatorname{diag}\left(\frac{1}{d_1}, \frac{1}{d_2}, \dots, \frac{1}{d_n}\right).$$

Example

Let

$$D = \begin{pmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$$

Then

$$D^{-1} = \begin{pmatrix} \frac{1}{4} & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{5} \end{pmatrix}.$$

This inverse is easy to understand: each coordinate was scaled independently, so the inverse simply rescales each coordinate back.

If a diagonal entry is zero, one coordinate is collapsed to zero. That information cannot be recovered.

4.5.2 Triangular matrices

Triangular matrices are also easy to test. If T is triangular, then

$$\det T = \text{product of diagonal entries.}$$

Thus T is invertible if and only if every diagonal entry is non-zero.

Example

Let

$$T = \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & 4 \\ 0 & 0 & 6 \end{pmatrix}.$$

Since

$$\det T = 1 \cdot (-1) \cdot 6 = -6 \neq 0,$$

the matrix is invertible.

On the other hand,

$$S = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 0 & 5 \\ 0 & 0 & 3 \end{pmatrix}$$

has determinant

$$\det S = 2 \cdot 0 \cdot 3 = 0.$$

So S is not invertible.

This is not merely a shortcut. A zero on the diagonal of a triangular matrix means that one pivot is missing. Without all pivots, the matrix cannot be reduced to the identity matrix.

4.5.3 Repeated or dependent rows

If two rows of a matrix are equal, the determinant is zero. If one row is a multiple of another, the determinant is also zero. More generally, if one row can be built from the others, the determinant is zero.

Example

Consider

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 3 & 6 & 3 \\ 0 & 1 & 4 \end{pmatrix}.$$

The second row is three times the first row. Therefore the rows are dependent, and

$$\det A = 0.$$

So A is not invertible.

The interpretation is important. If one row repeats information already contained in another row, then the matrix does not contain enough independent information to be reversed.

4.5.4 Products of matrices

The determinant rule

$$\det(AB) = \det A \det B$$

has a direct consequence for invertibility.

Theorem

If A and B are invertible square matrices of the same size, then AB is invertible, and

$$(AB)^{-1} = B^{-1}A^{-1}.$$

The order in this formula is essential. To undo the product AB, we must undo the last action first.

Suppose a vector x is transformed by B and then by A:

$$x \longmapsto Bx \longmapsto A(Bx) = ABx.$$

To recover x, first undo A, and then undo B:

$$ABx \longmapsto Bx \longmapsto x.$$

Thus the inverse is

$$B^{-1}A^{-1}.$$

We can verify algebraically:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I.$$

Similarly,

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I.$$

4.5.5 Matrices satisfying special identities

Sometimes a matrix is invertible because it satisfies an identity that already reveals its inverse.

Example

Suppose a square matrix A satisfies

$$A^2 = I.$$

Then A is its own inverse, because

$$AA = I.$$

Therefore

$$A^{-1} = A.$$

Such a matrix is called an involution.

Example

Suppose a square matrix A satisfies

$$A^2 - 3A + 2I = 0.$$

Then

$$A^2 - 3A = -2I.$$

Factor the left side:

$$A(A - 3I) = -2I.$$

Multiplying by $-\frac{1}{2}$, we get

$$A\left(-\frac{1}{2}(A - 3I)\right) = I.$$

Therefore

$$A^{-1} = -\frac{1}{2}(A - 3I) = \frac{1}{2}(3I - A).$$

This argument finds the inverse without performing Gauss–Jordan elimination.

These examples show that invertibility is not only a computational topic. It is also a structural topic. Sometimes the matrix tells us how it can be undone.

Summary

To recognize invertibility, look for structure:

- diagonal matrices are invertible when no diagonal entry is zero,
- triangular matrices are invertible when all diagonal entries are non-zero,
- repeated or dependent rows imply non-invertibility,
- products of invertible matrices are invertible,
- special identities may reveal the inverse directly.

4.6 Matrix equations

One of the main reasons for studying inverse matrices is that they allow us to solve matrix equations. The idea is similar to solving ordinary equations, but there is one crucial difference: matrix multiplication is not commutative. Therefore the side on which we multiply by an inverse matters.

4.6.1 Equations of the form $AX = B$

Suppose

$$AX = B,$$

where A is a known square matrix, B is a known matrix, and X is unknown. If A is invertible, we multiply on the left by A^{-1}:

$$A^{-1}AX = A^{-1}B.$$

Since $A^{-1}A = I$, this gives

$$X = A^{-1}B.$$

The inverse must be placed on the left because A is on the left of X. Writing XA^{-1} would not undo AX.

Example

Solve

$$AX = B,$$

where

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}, \qquad B = \begin{pmatrix} 4 & 1 \\ 5 & 2 \end{pmatrix}.$$

From an earlier example,

$$A^{-1} = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix}.$$

Therefore

$$X = A^{-1}B = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix} \begin{pmatrix} 4 & 1 \\ 5 & 2 \end{pmatrix}.$$

Compute the product:

$$X = \begin{pmatrix} 28 - 10 & 7 - 4 \\ -12 + 5 & -3 + 2 \end{pmatrix} = \begin{pmatrix} 18 & 3 \\ -7 & -1 \end{pmatrix}.$$

Thus

$$X = \begin{pmatrix} 18 & 3 \\ -7 & -1 \end{pmatrix}.$$

4.6.2 Equations of the form $XA = B$

Now suppose

$$XA = B.$$

Here A is on the right of X. To undo multiplication by A, we must multiply on the right by A^{-1}:

$$XAA^{-1} = BA^{-1}.$$

Since $AA^{-1} = I$, we get

$$X = BA^{-1}.$$

Example

Solve

$$XA = B,$$

where

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}, \qquad B = \begin{pmatrix} 4 & 1 \\ 5 & 2 \end{pmatrix}.$$

Again,

$$A^{-1} = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix}.$$

But now

$$X = BA^{-1} = \begin{pmatrix} 4 & 1 \\ 5 & 2 \end{pmatrix} \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix}.$$

Thus

$$X = \begin{pmatrix} 28-3 & -8+1 \\ 35-6 & -10+2 \end{pmatrix} = \begin{pmatrix} 25 & -7 \\ 29 & -8 \end{pmatrix}.$$

This is not the same answer as in the previous example. The equation is different because the unknown matrix is multiplied by A on a different side.

4.6.3 Equations with matrices on both sides

Some equations have the unknown matrix between two known matrices:

$$AXB = C.$$

If A and B are invertible, then we undo A on the left and B on the right:

$$A^{-1}(AXB)B^{-1} = A^{-1}CB^{-1}.$$

Therefore

$$X = A^{-1}CB^{-1}.$$

The order is not optional. Each inverse must be placed on the side where it can cancel the corresponding matrix.

4.6.4 Solving vector equations

The equation

$$Ax = b$$

where x and b are column vectors is a special case of $AX = B$. If A is invertible, then

$$x = A^{-1}b.$$

This gives a conceptual method for solving linear systems. In the next chapter we will study systems of linear equations in detail, and we will compare different methods: elimination, inverse matrices, and Cramer's rule.

Example

Solve

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

The determinant is

$$\det A = 2 \cdot 1 - 1 \cdot 1 = 1,$$

so

$$A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

Therefore

$$\begin{pmatrix} x \\ y \end{pmatrix} = A^{-1}b = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 5-3 \\ -5+6 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Thus

$$x = 2, \qquad y = 1.$$

4.6.5 Common mistakes

The most common mistake is to move matrices as if they were numbers. For example, from

$$AX = B$$

one may be tempted to write

$$X = BA^{-1}.$$

This is wrong in general. Multiplying by A^{-1} on the right does not cancel the A on the left of X. The correct step is

$$A^{-1}AX = A^{-1}B,$$

so

$$X = A^{-1}B.$$

Summary

When solving matrix equations:

- identify where the unknown matrix is placed,
- multiply by inverses on the side where cancellation is valid,
- preserve the order of multiplication,
- check that the matrices involved are invertible,
- verify the final answer by substitution when possible.

Inverse matrices are powerful because they allow equations to be solved by undoing operations. But the power comes with a responsibility: the order of multiplication must be respected at every step.

Chapter 5

Systems of Linear Equations

5.1 What systems of linear equations describe

This section will introduce systems of linear equations as mathematical objects that express several linear conditions at the same time. The aim will be to understand unknowns, equations, coefficients, and solution sets before choosing a method of solution.

5.2 Geometric meaning of solutions

This section will connect systems of linear equations with geometry. In two variables, equations describe lines; in three variables, they describe planes. The solution set is the set of points where all conditions are satisfied at once.

5.3 Gaussian elimination

This section will develop Gaussian elimination as a method of simplifying a system without changing its solution set. The focus will be on row operations, pivots, and the logic of reducing a complicated system to a readable one.

5.4 Cramer's rule

This section will introduce Cramer's rule as a determinant-based method for solving square systems with a non-zero determinant. The emphasis will be on what the rule says, when it applies, and why the condition $\det A \neq 0$ matters.

5.5 The inverse matrix method

This section will explain how a system written as $Ax = b$ can be solved by using A^{-1}, when the inverse exists. The method will be connected with the meaning of an inverse operation rather than presented as a mechanical trick.

5.6 Parametric systems and classification

This section will prepare the study of systems depending on a parameter. The goal will be to classify when a system has one solution, no solution, or infinitely many solutions, and to understand how this classification is reflected in algebraic and geometric structure.

Chapter 6

Vectors and Coordinates

This chapter is a placeholder for the first part of analytic geometry. It will develop the language of points, coordinates, and vectors before the detailed section structure is fixed.

Chapter 7

Lines

This chapter is a placeholder for the study of lines. It will cover lines in the plane and in space, with the detailed section structure to be decided later.

Chapter 8

Planes

This chapter is a placeholder for the study of planes. It will use the language of coordinates, vectors, and lines developed earlier, with the detailed section structure to be decided later.

Chapter 9

Sequences and Functions

This chapter is a placeholder for the introduction to sequences and functions. It will develop the basic language of functions, graphs, domains, values, and sequences before the detailed section structure is fixed.

Chapter 10

Limits and Continuity

This chapter is a placeholder for the study of limits and continuity. It will connect limits of sequences and functions with the notion of continuity and the behaviour of mathematical objects near a point.

Chapter 11

Derivatives and Function Analysis

This chapter is a placeholder for derivatives and their use in analysing functions. It will combine differentiation with monotonicity, extrema, convexity, asymptotes, and global understanding of function behaviour.

Chapter 12

Integrals

This chapter is a placeholder for integral calculus. It will develop antiderivatives, integration techniques, and the interpretation of integration as accumulation.

Chapter 13

Applications of Calculus

This chapter is a placeholder for applications of differential and integral calculus. It will focus on modelling, optimisation, geometry, motion, and other situations where calculus becomes a practical reasoning tool.